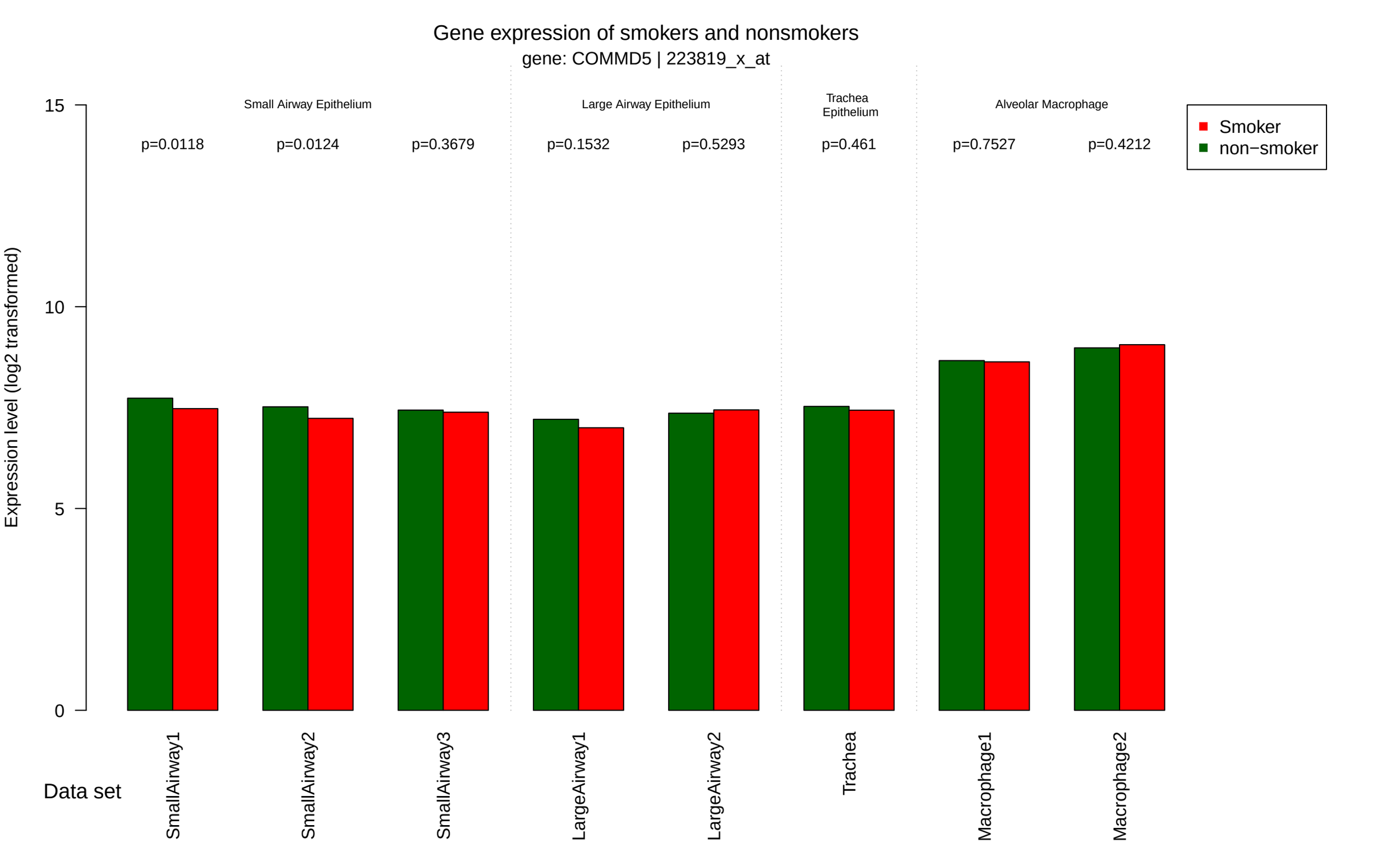
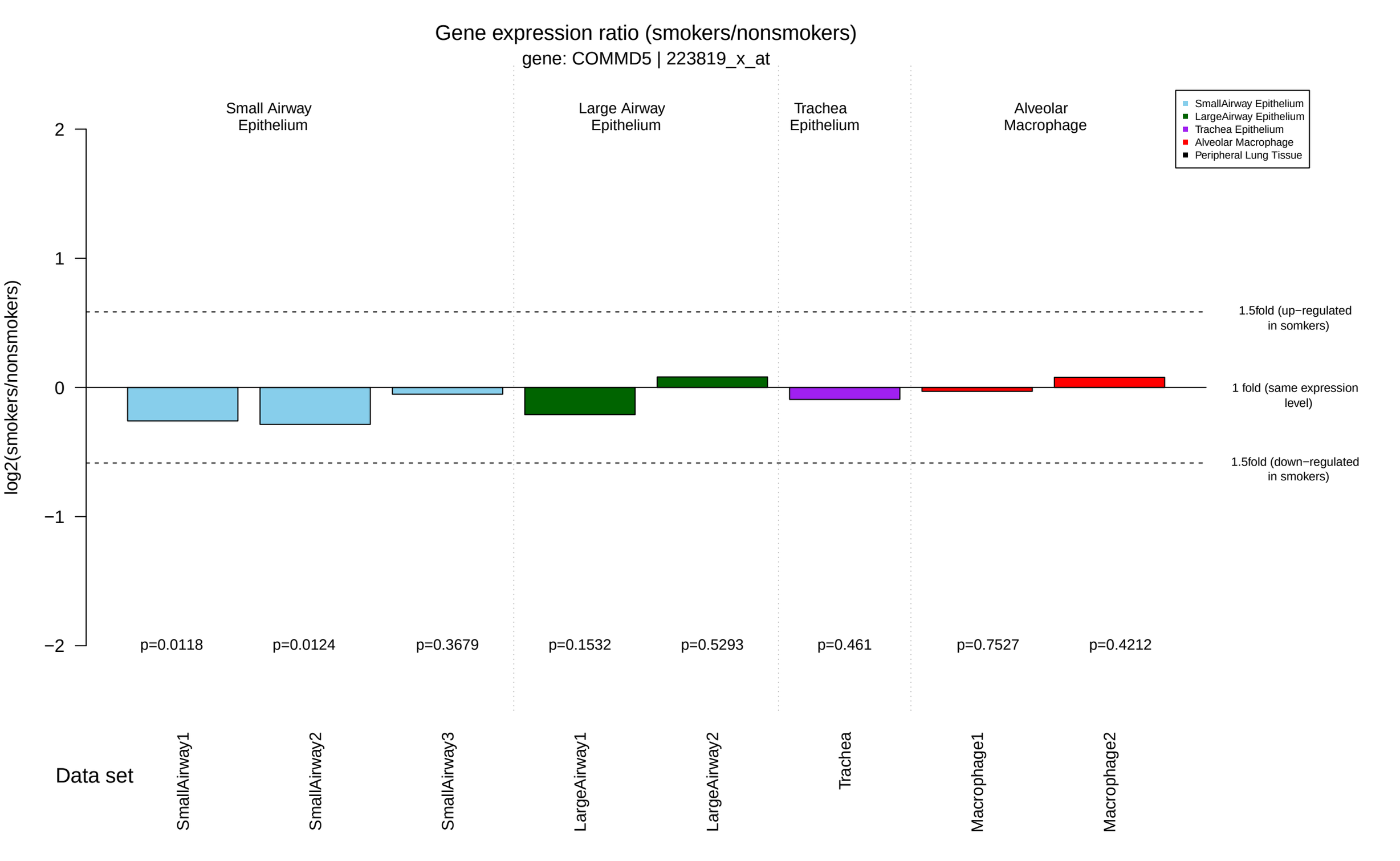
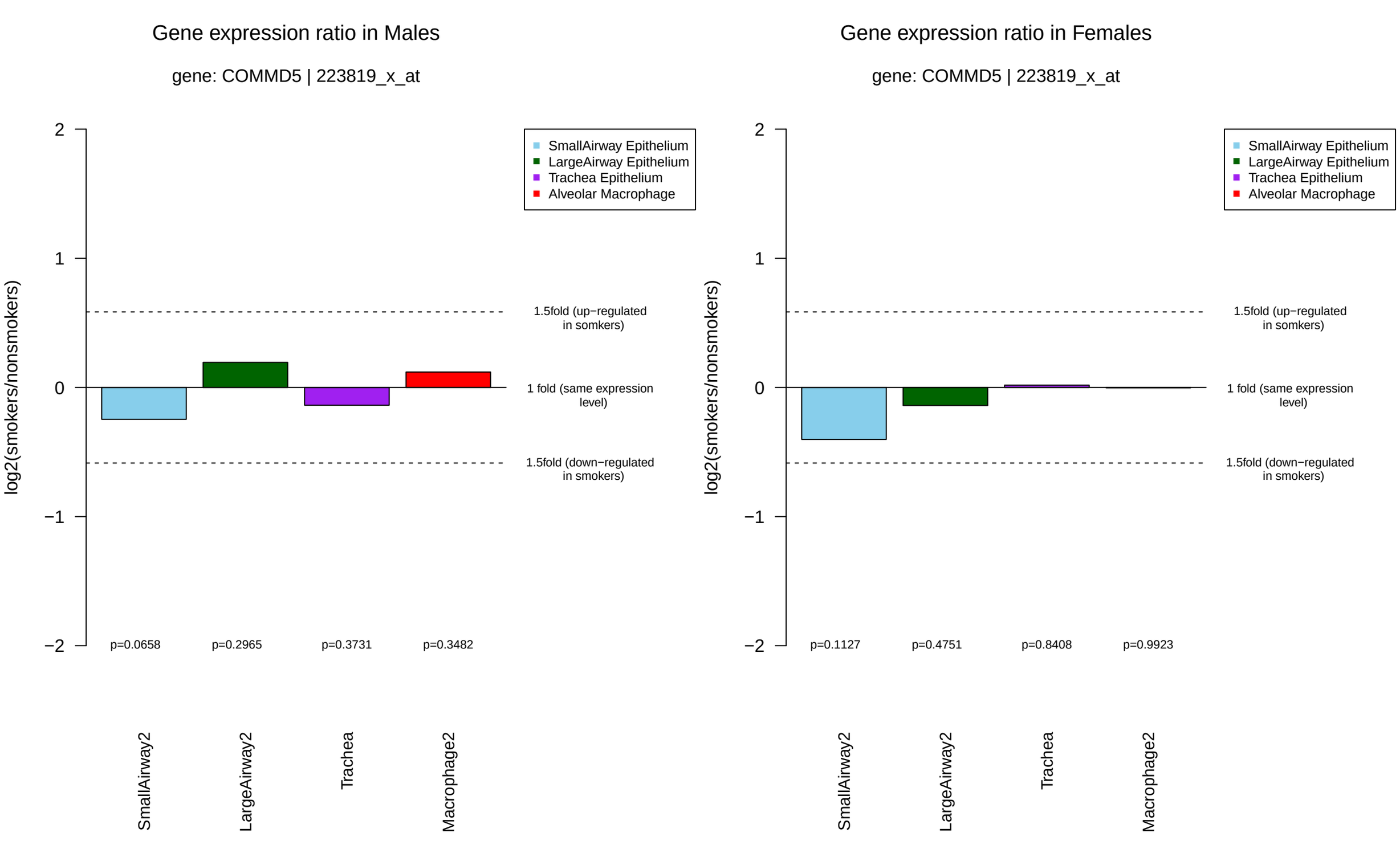
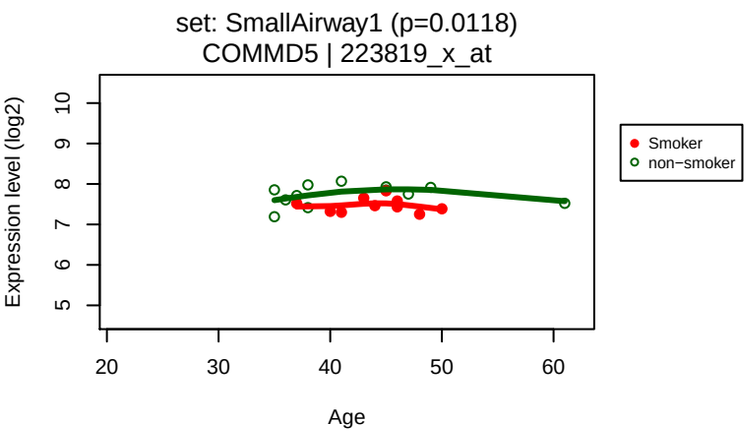
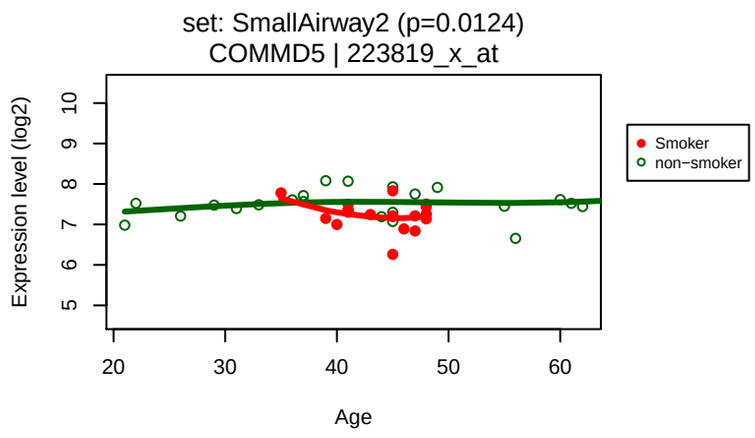
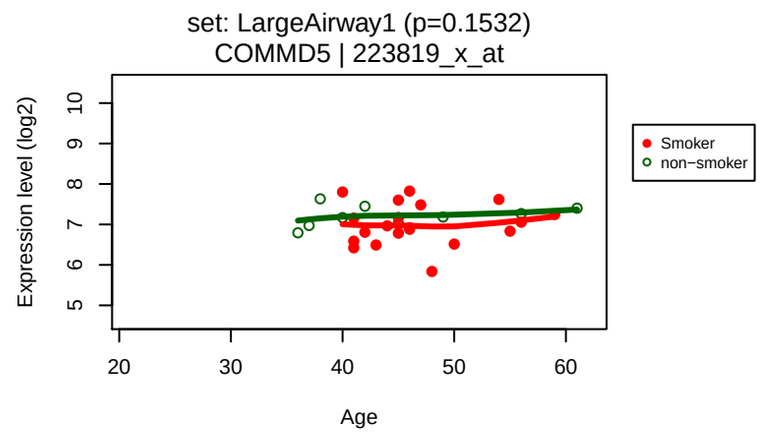

Gene expression of smokers and nonsmokers
gene: COMMD5 | 223819_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0118
p=0.0124
p=0.3679
p=0.1532
p=0.5293
p=0.461
p=0.7527
p=0.4212
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

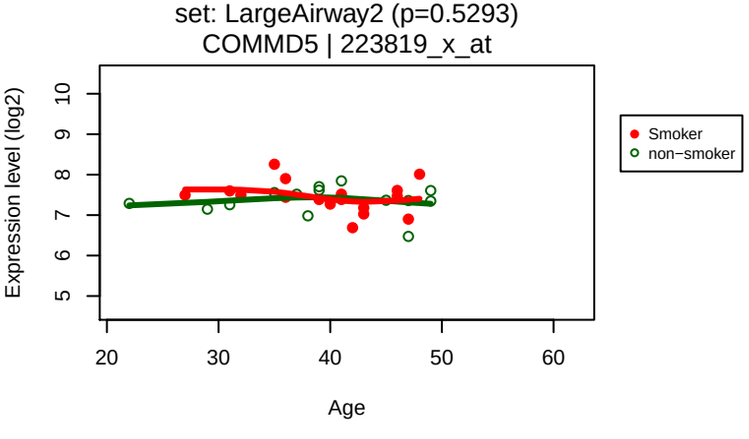

Gene expression ratio (smokers/nonsmokers)
gene: COMMD5 | 223819_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0118
p=0.0124
p=0.3679
p=0.1532
p=0.5293
p=0.461
p=0.7527
p=0.4212
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

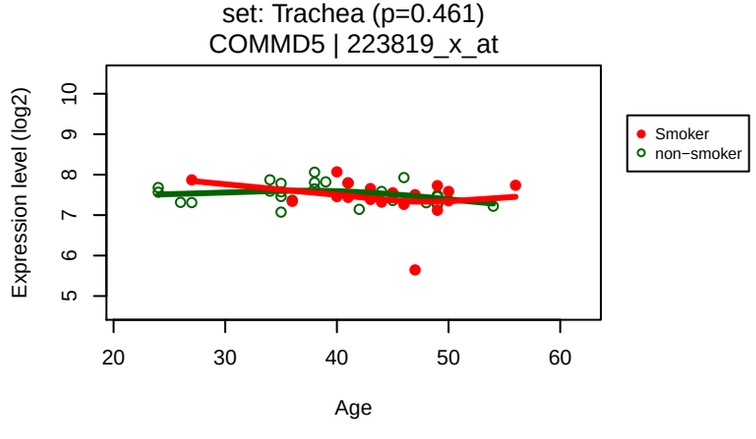

Gene expression ratio in Males
gene: COMMD5 | 223819_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0658
p=0.2965
p=0.3731
p=0.3482
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COMMD5 | 223819_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1127
p=0.4751
p=0.8408
p=0.9923
SmallAirway2
LargeAirway2
Trachea
Macrophage2

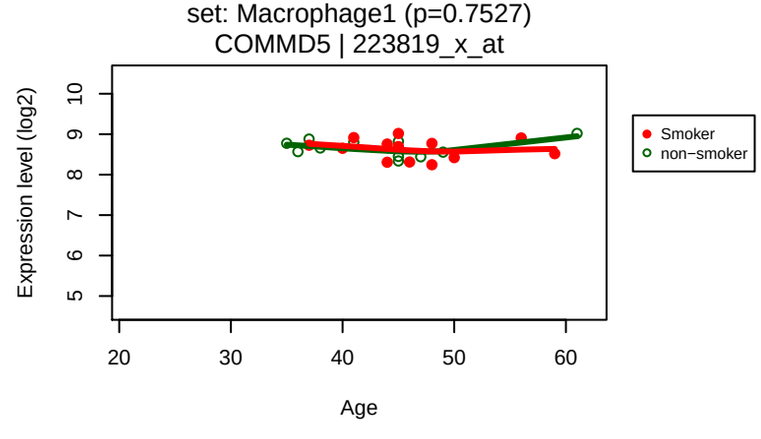

set: SmallAirway1 (p=0.0118)
COMMD5 | 223819_x_at
Expression level (log2)
Age
Smoker
non−smoker

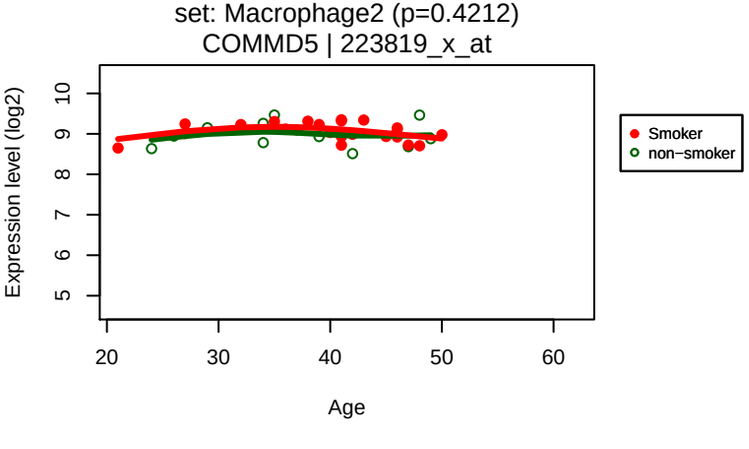

set: SmallAirway2 (p=0.0124)
COMMD5 | 223819_x_at
Expression level (log2)
Age
Smoker
non−smoker

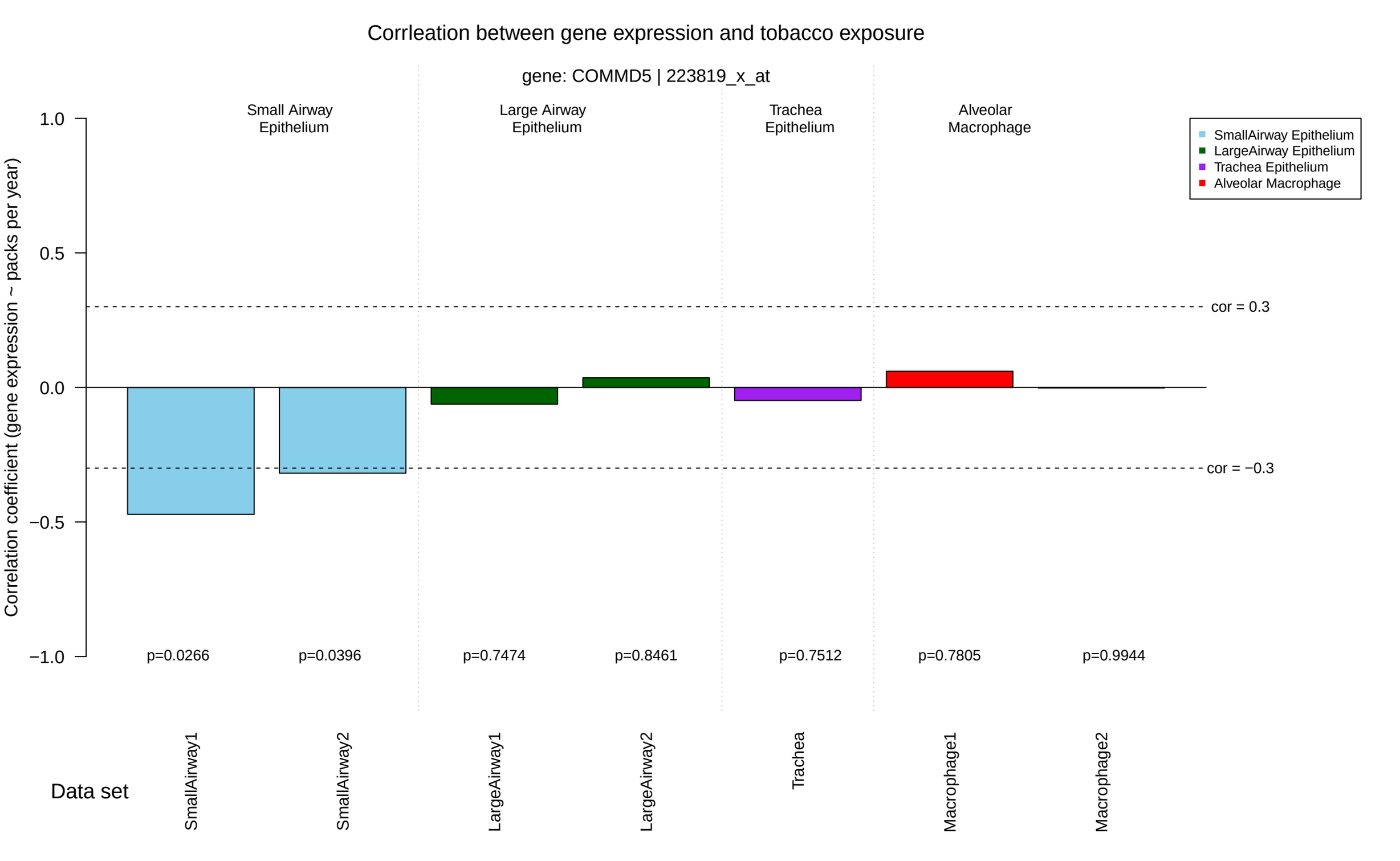

set: LargeAirway1 (p=0.1532)
COMMD5 | 223819_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5293)
COMMD5 | 223819_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.461)
COMMD5 | 223819_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7527)
COMMD5 | 223819_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4212)
COMMD5 | 223819_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COMMD5 | 223819_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0266
p=0.0396
p=0.7474
p=0.8461
p=0.7512
p=0.7805
p=0.9944
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2